6.4 Properties of the Definite Integral

Name _____ Period ____ Date _____

AP Calculus 6.4 · Student sheet

I can…

- use properties of the definite integral.
- find the average value of a function on an interval.
- find displacement and total distance from velocity.

Vocabulary · fill in the missing word

Average value — _____

Mean Value Theorem for Integrals — if f is _____ on [a, b], then f(c) equals the average value for some c in [a, b]

Total distance — ∫ |v(t)| dt, the distance traveled in both _____

Part 1 · Flatten the hill

$y = x^2$ on [0, 3]; the area under it is 9.

Question	Answer
Height of a rectangle on [0, 3] with area 9	
Where does x^2 equal that height?	

Part 2 · Notes and examples (fill in each note, then work the example)

Adjacent intervals: ∫ from a to b + ∫ from b to c = ∫ from a to _____. Reversing the limits changes the sign.

Example 1

$\int_0^2 f(x)\,dx = 5$ and $\int_2^5 f(x)\,dx = -3$. Find $\int_0^5 f(x)\,dx$ and $\int_5^2 f(x)\,dx$.

Average value: f_avg = _____ · ∫ from a to b of f(x) dx.

Example 2

Find the average value of $f(x) = 3x^2$ on [0, 2].

Mean Value Theorem for Integrals: f(c) = the _____ value for some c in [a, b].

Example 3

For $f(x) = 3x^2$ on [0, 2], find c with f(c) equal to the average value.

Motion: displacement = ∫ v(t) dt; total distance = ∫ _____ dt.

Example 4

$v(t) = t - 2\ \frac{\text{m}}{\text{s}}$ on $[0, 6]$. Find the displacement and the total distance.

Example 5

Evaluate $\int_0^6 |x - 2|\, dx$.

Part 3 · Partner practice

Partner 1

Find the average value of $f(x) = x^2$ on $[0, 3]$.

Partner 2

(AP® Practice) If $\int_1^5 f(x)\, dx = 8$ and $\int_1^3 f(x)\, dx = 3$, then $\int_3^5 f(x)\, dx =$

(A) 5 (B) 11 (C) −5 (D) 24